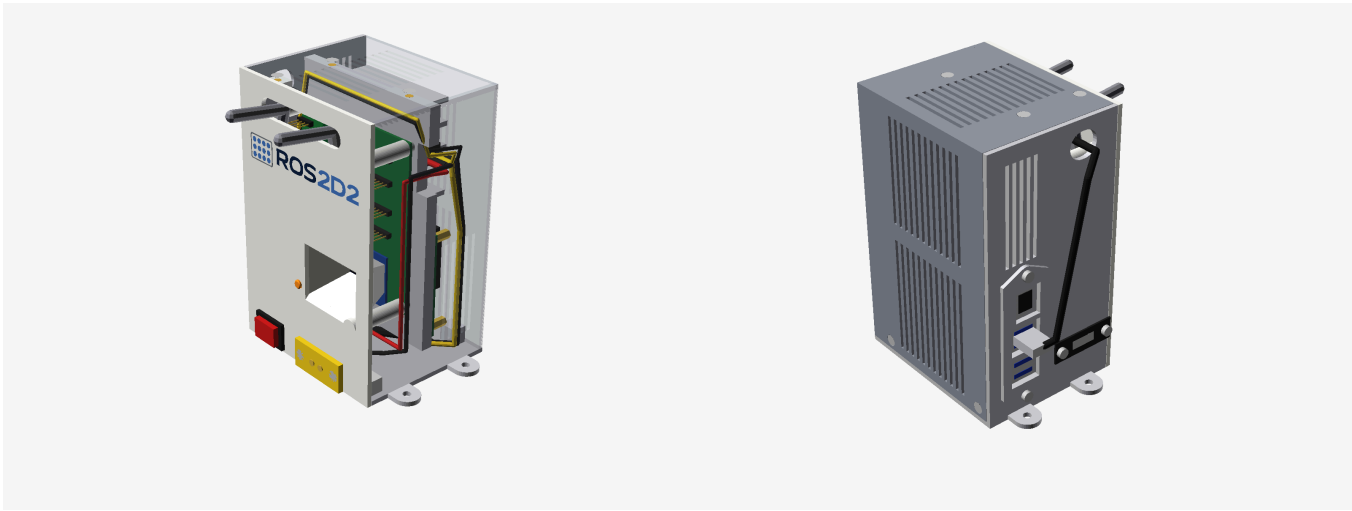
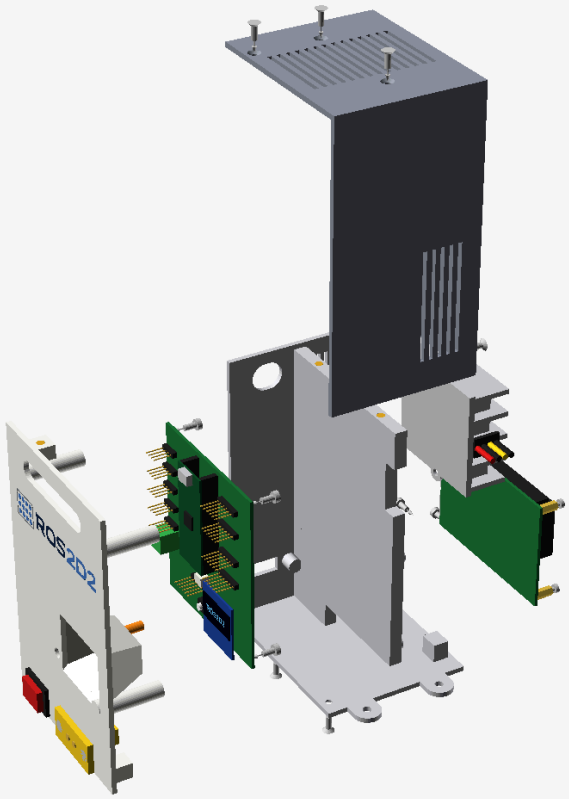
ROS2 rack box · Assembly instructions

Second box, rack style: the Pico board, the Pi and the converter stand on edge, side by side. 78 × 95 × 145 mm (1.07 L). The Pi sits low and the converter high, so its heat does not rise into the Pi. It takes a Pi 4 or a Pi 5: only the back frame changes. OLED, logo, switch and XT60 are on the main face; the Pi's USB and an HDMI panel jack at the back. Every screw goes into an insert. Before printing, measure what the README lists.



Exploded view


ROS2D2

Hardware and parts

Printed parts

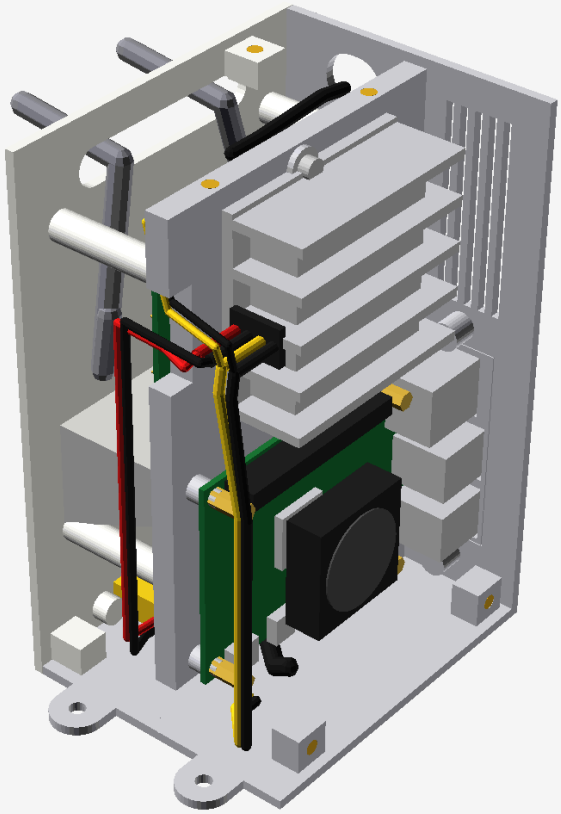
`pared_multicolor_bambu.3mf`	Main face with the logo in 4 colours (AMS: white, navy, blue, grey)
`pared_monocolor.stl`	or: single-colour main face, engraved logo
`base.stl`	Base: floor, back, central divider
`tapa.stl`	Shell (roof + side + narrow front)
`marco_pi4.stl`	Back frame for the Pi 4 (Ethernet at the top)
`marco_pi5.stl`	or: back frame for the Pi 5 (Ethernet at the bottom)
`pulsador.stl`	DISPLAY push rod

Qty	Part	Step
19×	M3 heat-set insert (5.7 mm long, 4.0 mm hole)	1
9×	M3 × 8 countersunk screw (ISO 10642)	4, 6, 7
8×	M3 × 6 button head screw (ISO 7380)	3, 5
2×	M3 × 6 socket head cap screw (DIN 912)	2
4×	M2.5 × 6 screw	3
4×	M4 × 12 screw (or what the R2's mount needs)	8
4×	M2.5 heat-set insert (5 mm long, 3.6 mm hole)	1

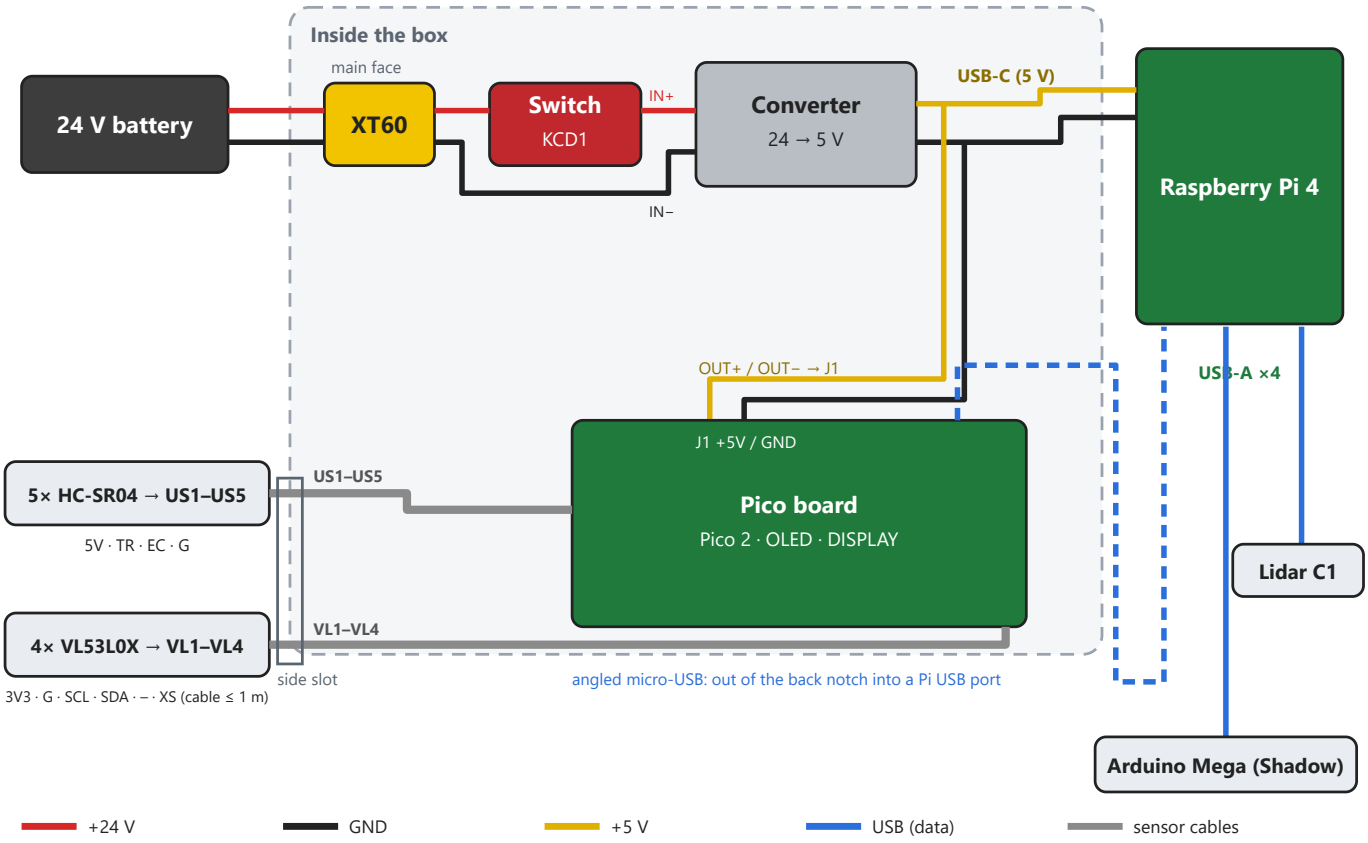
Other parts

1	KCD1 rocker switch (panel hole 19.2 × 13 mm)
1	XT60E-F panel connector (AMASS)
1	micro-USB 2.0 to USB-A cable, 90° side angle, 30 cm (see «Angled plugs»)
—	0.5 mm^2 wire for the 5 V to terminal J1, 1 mm^2 for the 24 V
1	Edge grommet strip for the 35 × 10 mm slot
1	USB-C male pigtail (5 V to the Pi), 90° up/down angle, 15–20 cm; 3 A for a Pi 4, 5 A for a Pi 5
1	Angled 90° micro-HDMI male to HDMI female panel-mount cable with 2 M3 screws, 30 cm (measure its flange: `hdmi_*`)

Inside (no shell)



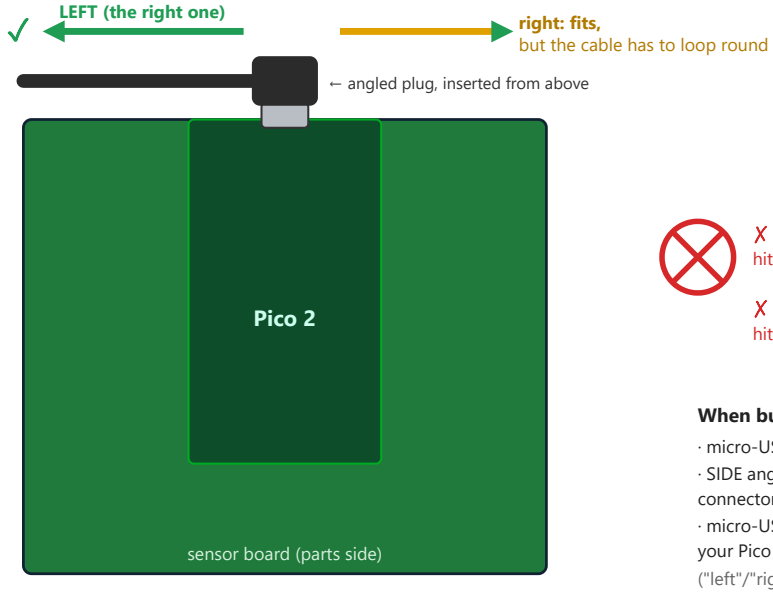
Wiring



Angled plugs: which way the cable goes

1 · Angled micro-USB (Pico → Pi) · both boxes

Looking at the Pico's parts side, its USB at the top (as it sits on the sensor board):



✗ **towards you (the parts side):**
hits the sensors' Dupont plugs

✗ **backwards (behind the board):**
hits the divider in box C

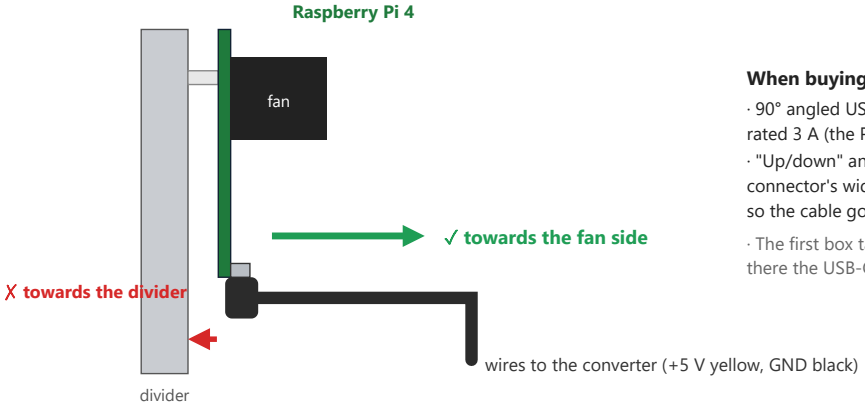
When buying:

· micro-USB 2.0 (not 3.0), 30 cm (fits both boxes).
· SIDE angle: the cable leaves sideways, along the connector's long side (not up or down).
· micro-USB only goes in one way: compare the photo with your Pico and check the cable will end up on the LEFT.
("left"/"right" mean different things to different brands)

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2 · Angled USB-C pigtail (converter → Pi) · box C

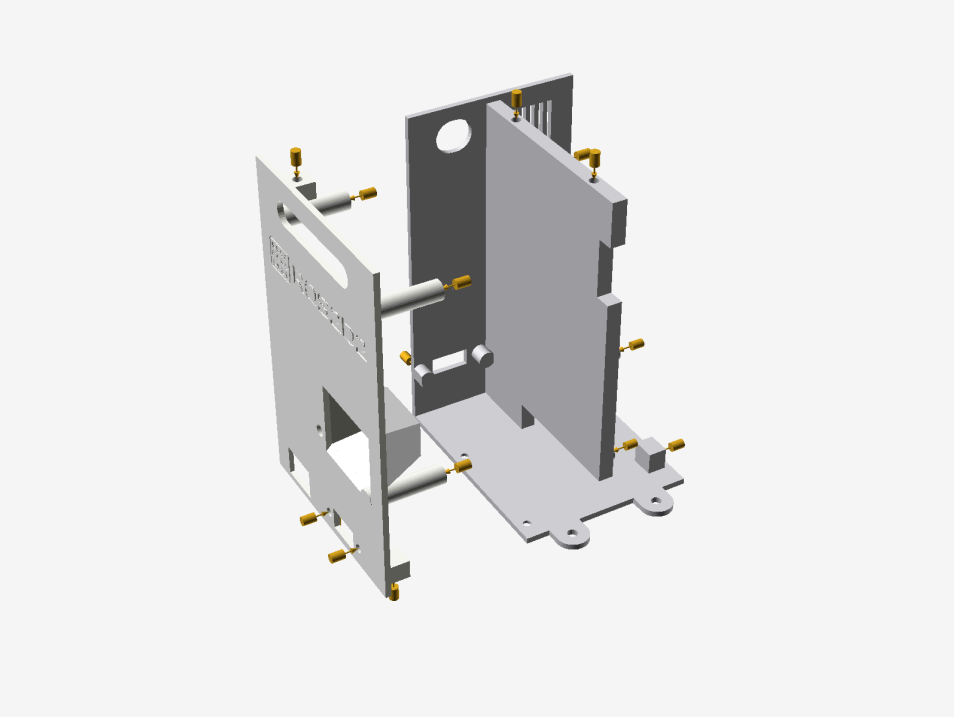
The Pi seen from the narrow front: standing, parts and fan to the right, USB-C at the bottom:



When buying:

· 90° angled USB-C male pigtail (loose wires), rated 3 A (the Pi 4 wants 5 V / 3 A), 15–20 cm.
· "Up/down" angle: the cable leaves square to the connector's wide face. USB-C is reversible: turn it so the cable goes towards the fan.
· The first box takes the same cable (or a straight one): there the USB-C faces the roof, cable towards the middle.

1 Heat-set inserts

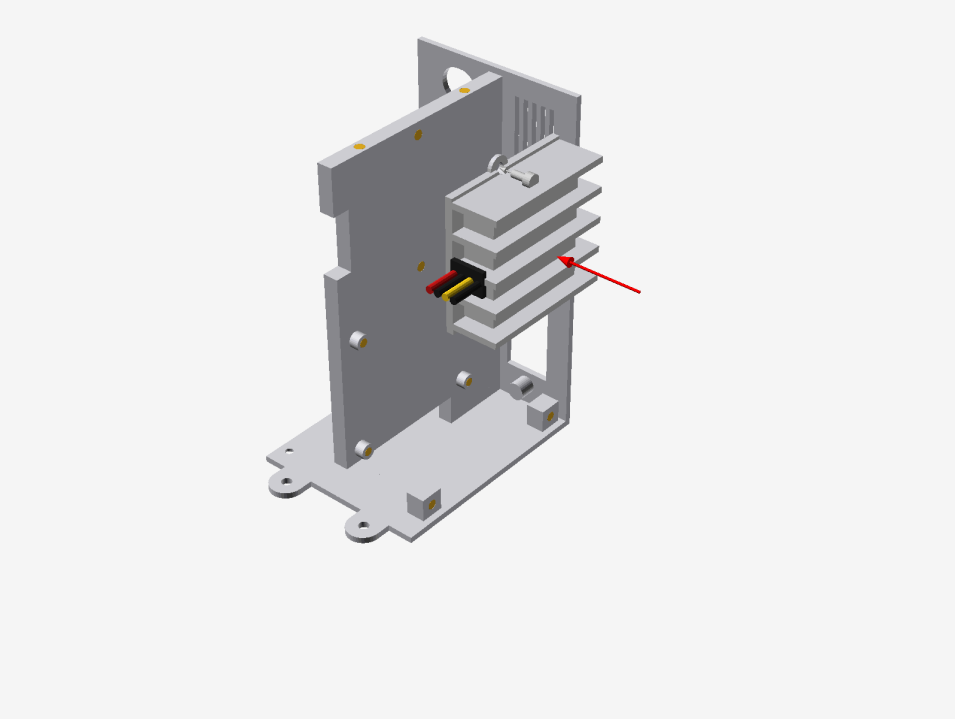


19× M3 insert

4× M2.5 insert

With the iron at about 220 °C press the inserts in straight and flush. On the main face, 7 M3: 4 in the columns, 2 in the bottom tabs and 1 in the top tab; plus 2 M3 from outside in the XT60's holes. On the base, 10 M3: 2 on top of the divider, 2 in the divider for the converter, 2 in the floor blocks and 4 from outside the back (2 for the frame, 2 for the HDMI); and the 4 M2.5 in the Pi's standoffs.

2 Converter onto the divider, high

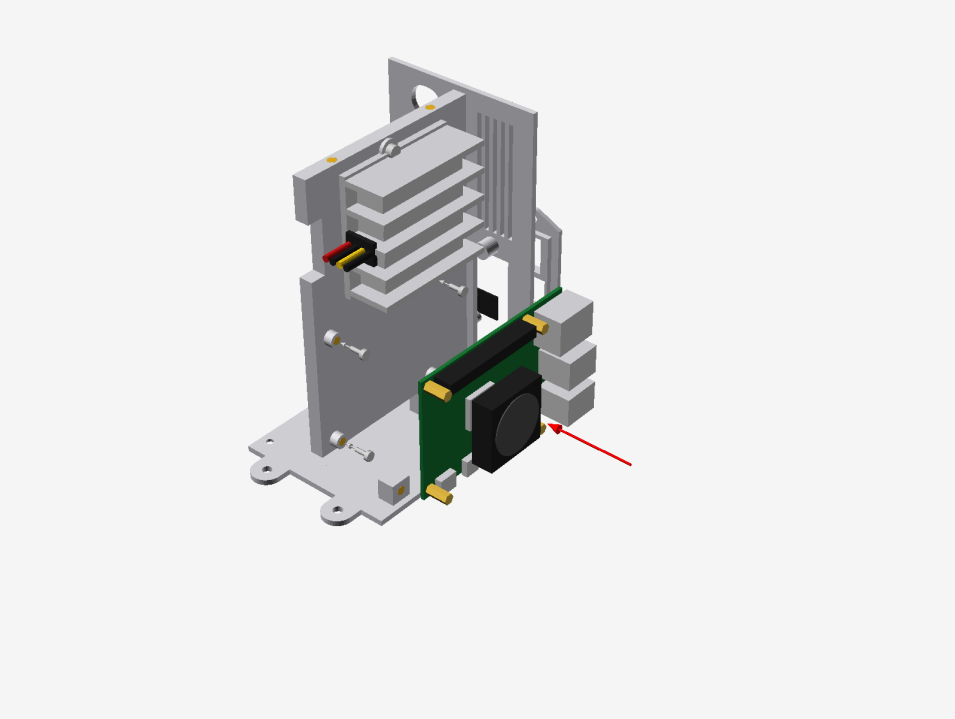


2× M3 × 6 socket head

On edge, at the very top, its base plate against the divider, fins outwards, wires to the front: they go through the top pass in the divider to the Pico side. Its heat rises out of the roof without passing the Pi. Check its holes (`conv_holes`) with a caliper first.

3

Raspberry Pi, frame and HDMI



4× M2.5 × 6 4× M3 × 6

Pi 4 or Pi 5 on its 4 standoffs, low down, USB and Ethernet in the back window, USB-C at the bottom. Plug in the angled USB-C (5 V from the converter, down in front of the Pi) and the angled micro-HDMI into HDMI 0; that cable runs under the Pi through the divider's bottom pass. Screw on, from outside, the frame for your Pi (`marco_pi4` or `marco_pi5`) and the HDMI panel jack.

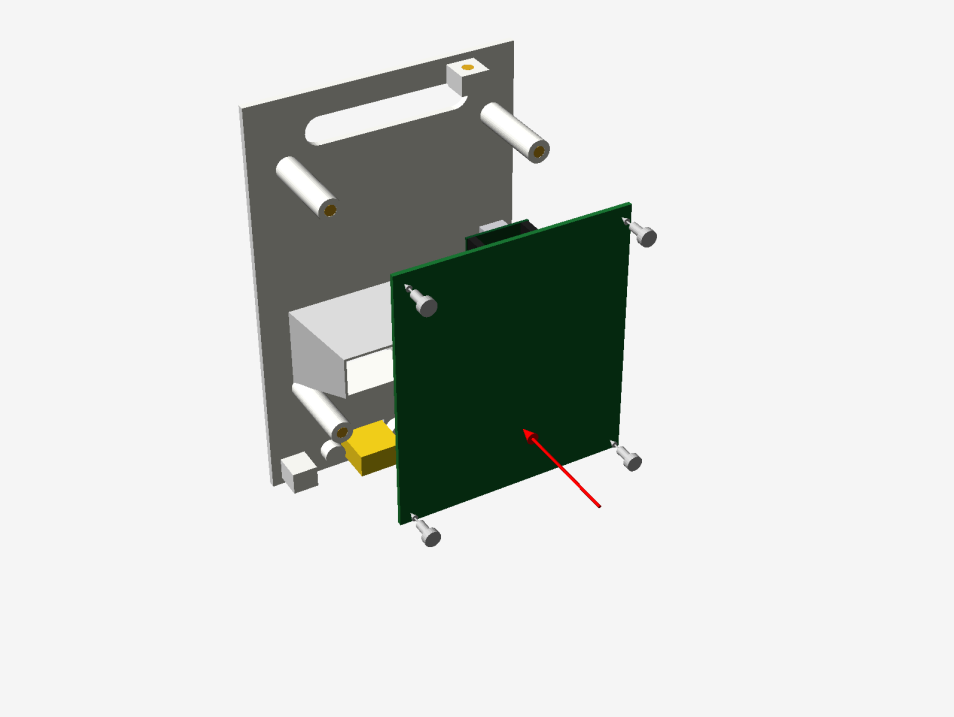
4 XT60 and switch on the main face



2× M3 × 8 countersunk

From outside, low down: the XT60E-F turned on its side with 2 screws into the inserts, and the switch snapped in.

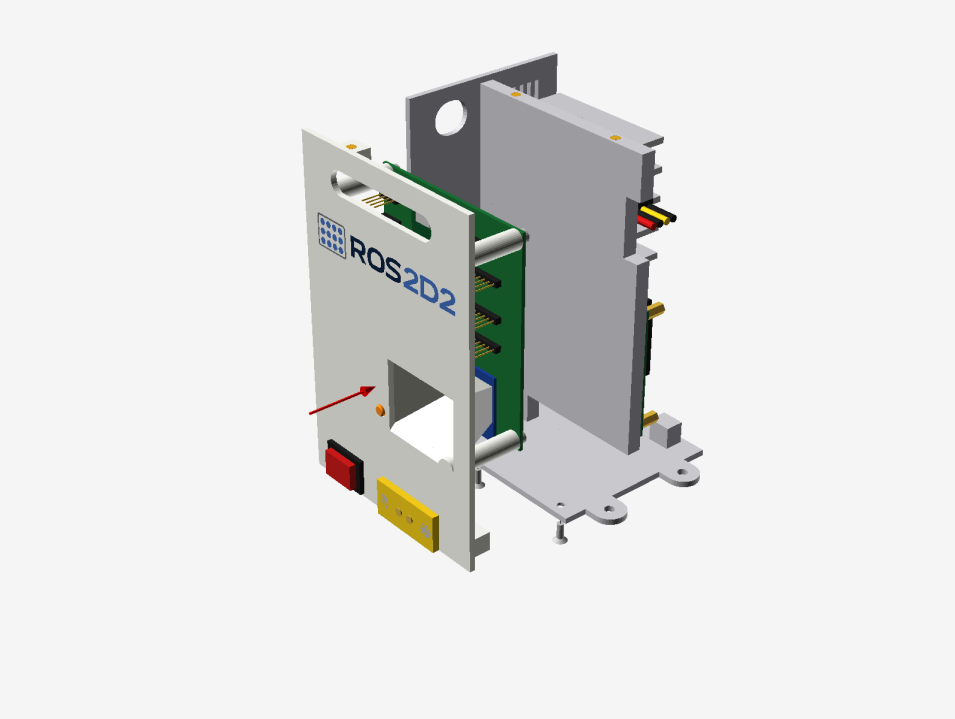
5 Push rod and Pico board



4× M3 × 6

Slide the rod into its guide from inside and screw the board onto the 4 columns, parts towards the wall. Tighten the 5 V in terminal J1, plug the angled micro-USB into the Pico and connect the sensors: their cables leave through the slot at the top.

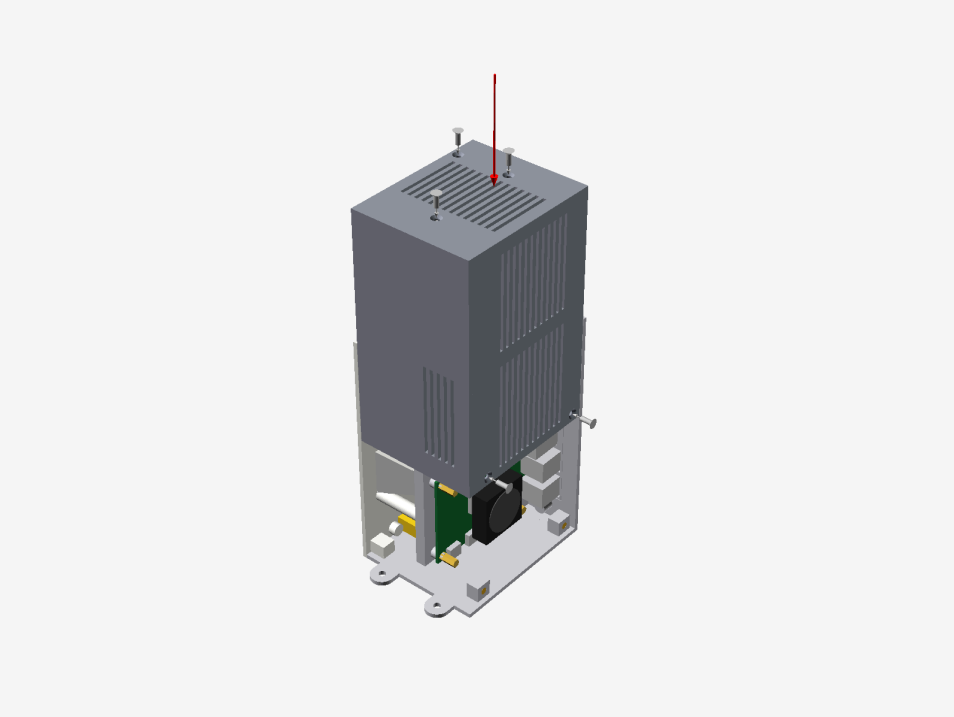
6 Main face onto the base



2× M3 × 8 countersunk

The 24 V (XT60 → switch → converter) run up in front of the Pico board and cross through the divider's top pass; the 5 V to terminal J1 come back through the same pass and over the board. Screw the face on from under the floor.

7 Shell



5× M3 × 8 countersunk

Lower it from above and fit the 3 roof screws and the 2 side screws. Air comes in through the low vents (front and side, by the Pi) and leaves through the high ones and the roof.

8 USB and onto the R2



4× M4 × 12

Bring the micro-USB out of the notch in the back into a Pi USB port. Screw the box to the R2 through its 4 ears.